

Cell Biology Basics

1. What is a cell?

A cell is the smallest unit of life. Every living organism is made of one or more cells. Cells carry out the chemical reactions that keep an organism alive, and they can copy themselves by dividing.

2. Parts of a cell

The cell membrane surrounds the cell and controls what enters and leaves it. The nucleus stores the genetic instructions called DNA. Mitochondria release energy from food in a process called respiration. The cytoplasm is the jelly-like fluid that fills the cell and holds the other parts in place.

3. Plant cells and animal cells

Plant cells have three parts that animal cells do not have: a rigid cell wall made of cellulose, a large central vacuole filled with sap, and chloroplasts that capture light for photosynthesis.